

Medical Image Segmentation

Overview

This project implements a U-Net architecture from scratch to perform retinal blood vessel segmentation on medical fundus images. The goal is to accurately identify and segment blood vessels in retinal images

This project focuses on understanding:

- How CNNs work and how they are used for image segmentation.
- How to train/test segmentation models and compute Dice and IoU scores for performance evaluation.

Dependencies

Install the required packages: Go to the project folder and enter this to download all packages

- `pip install -r requirements.txt`

File Directory

- demo.py – Contains the function code to run and test the two models. The demo function helps provide visual representation of the model prediction compared to the ground truth
- RetinaDataset.py – Contains a class which sets up data and normalizes it for training.
- helper.py – Contains the function code of the loss functions and to compute metrics to evaluate and compare different model performance.
- UNet_2blocks.py and UNet_3blocks.py – Contains the class for each neural network which includes architecture, training and testing functions for each.
- U2_net.pth and U3_net.pth - Is the model state saved of the two models created
- image/ - folder containing images of all demos and model training iteration.

How to Run

- The `test_model()` function in demo.py displays a menu that allows the user to select one of the available trained models. Once a model is chosen, the method loads the corresponding saved weights, evaluates the model on the testing dataset, and reports key performance metrics such as Dice score, IoU, and accuracy. Additionally, it visualizes the model's predictions on the first four images from the test dataset for qualitative comparison.